

12/2/26

Convolution in Radio Astronomy

$x(t)$ → a sequence (can be voltage, fields, probabilities ...)

$y(t)$ → another sequence

Then the convolution of $x(t), y(t)$ is

$$(x * y)(t) = \int_{-\infty}^{\infty} x(\tau) \cdot y(t - \tau) d\tau$$

τ is a dummy variable that disappears after the evaluation of the convolution integral.

This can be discretized as:

$$(x * y)[n] = \sum_{\tau=-\infty}^{\infty} x(\tau) y(n - \tau)$$

$x(\tau)$ → function at an instant (τ).

$y(n - \tau)$ → n → Samples that we have, $(-\tau)$ → flips the time order and $(n - \tau)$ on the whole shifts it behind by 'n' so that the last sample of $y(n - \tau)$ is aligned with the first sample

The Integration (or) Summation captures all possible "delay" values, enabling us to find the "effect" of all τ 's on the sequence.

In effect

- ① $x(\tau)$ is scaled by $y(T-\tau)$
- ② This is done for different (τ) 's
- ③ All the scaled values at different (τ) 's are added.

Example

The time reversal happens because in ~~real~~ reality if $x(n)$ represented a linear time invariant system, then as time progresses, $y(n)$'s first sample will arrive at $x(n)$'s first sample.

$$\begin{array}{cccc} x(0) & x(1) & x(2) & x(3) \\ y(3) & y(2) & y(1) & y(0) \end{array} \quad \left. \vphantom{\begin{array}{cccc} x(0) & x(1) & x(2) & x(3) \\ y(3) & y(2) & y(1) & y(0) \end{array}} \right\} t=0$$

$$\begin{array}{cccc} x(0) & x(1) & x(2) & x(3) \\ y(3) & y(2) & y(1) & y(0) \end{array} \quad \left. \vphantom{\begin{array}{cccc} x(0) & x(1) & x(2) & x(3) \\ y(3) & y(2) & y(1) & y(0) \end{array}} \right\} t=1$$

Now the "response" of $x(n)$ to $y(n)$ is what is obtained by summing at different

delays and adding.

∴ Convolution can be used to obtain the response of a ~~linear time invariant system~~ LTI system to a signal.

(or)

How can a signal change when it encounters an LTI system

Antennas, Amplifiers, filters etc are usually considered to be LTI

Example

$$x: [1, 2, 3]$$

$$y: [4, 5, 6]$$

$$\begin{array}{ccc|ccc} & & & 1 & 2 & 3 & 1 & 2 & 3 \\ & & & 4 & 5 & 6 & 4 & 5 & 6 \\ 4 & 5 & \textcircled{6} & & & & & & \end{array} \quad \left. \begin{array}{l} \\ \\ g(0) = 6 \end{array} \right\} \begin{array}{l} \\ \\ g(1) = 5 + 12 = 17 \end{array} \quad \left| \begin{array}{l} \\ \\ g(2) = 4 + 10 + 18 = 32 \end{array} \right.$$

~~g(0)~~

$$\begin{array}{ccc|ccc} & 1 & 2 & 3 & & & 1 & 2 & 3 \\ & 4 & 5 & 6 & & & & 4 & 5 & 6 \\ \hline & & & & & & & & & \end{array}$$

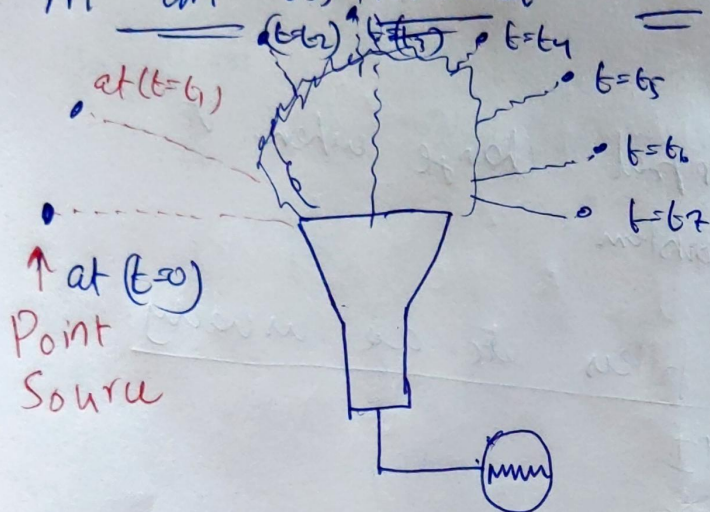
$$g(3): \quad 8 + 15 = 23 \quad \left| \quad g(4) = 12 \right.$$

$$\therefore x * y = [6, 17, 32, 23, 12]$$

length of $x \times y$ is longer than both x & y

$$L(x \times y) = \underline{L(x) + L(y) - 1}$$

In an astronomical scenario

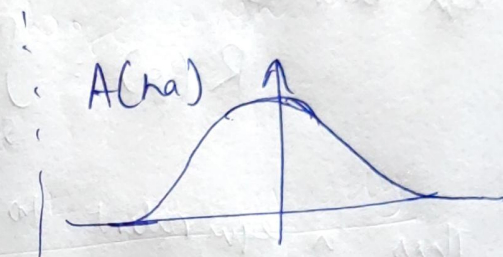
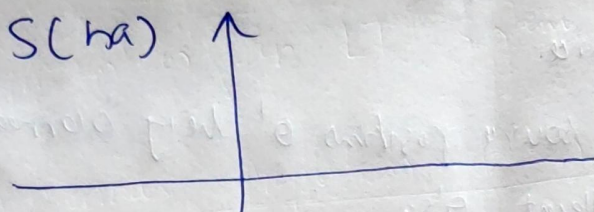


The source has a constant brightness of 1, and its extent is infinitesimally small. A telescope placed on the earth will scan across this as the earth rotates. We will consider a transit telescope [telescope is not "tracking" the source]. This implies as time changes, the source will be at different hour angles. As the source rises, the HA is $-ve$, at meridian it will be (0) & positive as it moves to the west.

$$\therefore S(ha=0) = 1$$

The antenna response at that ~~is that~~ ^{hour angle} will ~~be~~ ^{scale the source} ~~the~~ ^{flux} brightness/flux.

$\Rightarrow$ if the antenna response was gaussian then the ~~antenna~~ response will be like this:



The antenna response is fixed in time, which means the ~~the~~ max of the resp. is always pointing up.

$\therefore$ The source "slides" over the antenna response

$\Rightarrow$ The source ~~the~~ brightness distribution is involved with the antenna's response.

Mathematically: as the earth rotates and the antenna max makes different angles with source position, the response R becomes,

$$R \propto A(\theta' - \theta) S(\theta)$$

where θ' is the angle antenna makes with the source ~~in~~ source

$\Rightarrow$ As we assumed a pt. source, we will measure only the antenna response itself. This is called the "point spread function".

If the source was extended the the convolution eqn becomes

$$R(\theta) = \int_{\text{Source}} A(\theta - \theta') \cdot I(\theta') d\theta'$$

This is equivalent to source positions θ' being "smeared" by the antenna at different θ 's.

this principle is used for measuring antenna responses in "real experiments".

Question

Atmosphere / Ionosphere can sometimes cause random fluctuations in the brightness. How will this be affected by the system noise, which is randomly distributed? What will be the resultant variance, mean?

13/2/26

Fourier Transform, Auto & Cross-Correlation

→ Used to find the signal ^{frequency} components in a system.

→ Can be in ~~the~~ ^{temporal} frequency, spatial frequency space. These two are of interest to us.

time series → Temporal
Images → Spatial

Fourier transform

Suppose a complex function, $\hat{f}(\xi) = A e^{i\theta}$. Multiplication of this with a "kernel" $e^{i2\pi kx}$ will give us:

$$\hat{f}(\xi) e^{i2\pi kx} = A e^{i\theta} \cdot e^{i2\pi kx} \\ = A e^{i(2\pi kx + \theta)}$$

Amplitude of above $\Rightarrow |A|^2$

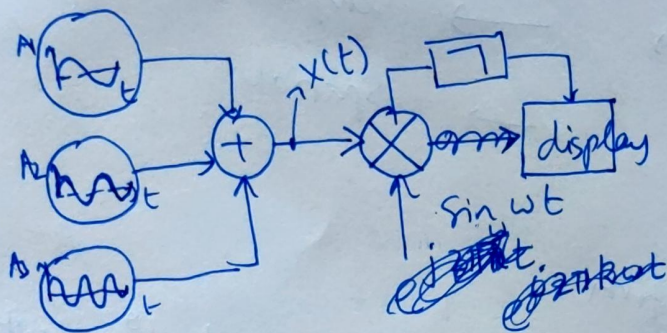
Phase $\rightarrow \tan^{-1} \frac{\text{Im}[A e^{i(2\pi kx + \theta)}]}{\text{Re}[A e^{i(2\pi kx + \theta)}]}$

This is sort of like applying a very sharp filter,

~~Says~~ $\Rightarrow$ only if θ has a freq component that has a match at $2\pi k$ will produce a non zero amplitude

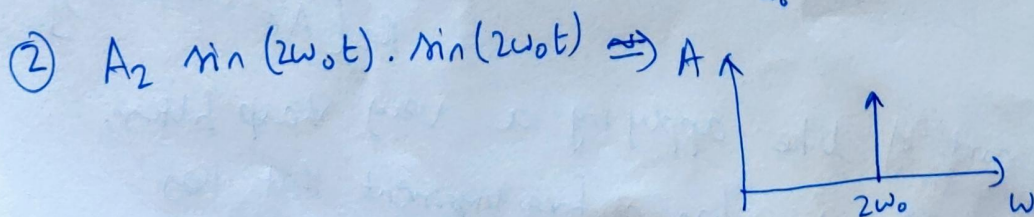
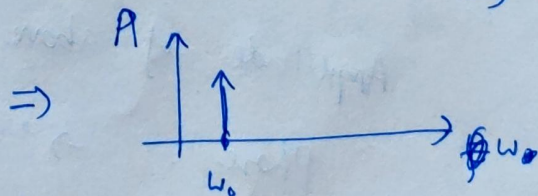
Detour: Locke in detection uses this technique.

Now if we want to measure other frequencies, then I just have to "tune" the kernel such that I'm able to "pick" out the signals show up with a finite amplitude, like below:



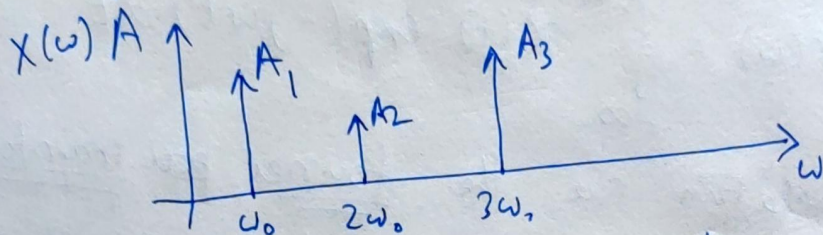
$$\begin{aligned}
 ① &= A_1 \sin(\omega t) \\
 ② &= A_2 \sin(2\omega t) \\
 ③ &= A_3 \sin(3\omega t)
 \end{aligned}
 \quad
 \begin{array}{l}
 \times \sin(\omega t) \\
 \times \sin(2\omega t) \\
 \times \sin(3\omega t)
 \end{array}
 \quad
 \begin{array}{l}
 \sin(\omega t) \\
 \sin(\omega t) \\
 \sin(\omega t)
 \end{array}$$

$$① \quad A_1 \sin(\omega_0 t) \cdot \sin(\omega_0 t) = \cos(\omega_0 - \omega_0)t + \cos(\omega_0 + \omega_0)t \xrightarrow{\quad} \rightarrow 0.$$



Similarly for $3\omega_0$ also.

∴ If I increased the persistence to ∞ in my display, and am able to track the ~~initial~~ frequency of the "Local oscillator", then I can get a spectra like this:



I can add another channel with $\cos(\omega t)$ to obtain both sine & cosine components. This will let me construct both ~~the~~ amplitude & phase vs frequency displays.

But

- This assumes the signal lasts for ∞
- ~~And is periodic~~ ^{but based on this,} ~~that is:~~ This signal could be a periodic & exist only for my meas window.

$$X(t) \propto \left[A_1 \sin(\omega_0 t) \cdot e^{+j2\pi\omega_0 t} + A_2 \sin(2\omega_0 t) \cdot e^{+j2\pi(2\omega_0)t} + A_3 \sin(3\omega_0 t) \cdot e^{+j2\pi(3\omega_0)t} \right]$$

If instead of choosing discrete steps, I sweep all frequencies, (remember, the signal here lasts until ∞)

Then,

~~$\Rightarrow x(t) =$~~

~~$x(t) =$~~
$$x(t) = \int_{-\infty}^{\infty} X(\omega) \exp(j\omega t) \cdot d\omega$$

From here, an inverse relation can be written: $X(\omega) \sim A_1(\omega_0) + A_2(2\omega_0) + A_3(3\omega_0)$

$$X(\omega) = \int_{-\infty}^{\infty} x(t) \exp(-j\omega t) dt.$$

These two relations are the fourier pair transforms.

$x(t)$ is built by functions at $\omega_0, 2\omega_0, 3\omega_0$.

while

$X(\omega)$ is what we get when we take $x(t)$,
~~multiply~~ apply a sharp, infinitesimally wide
filter.

In reality, we do this integration over a time "window"
and on sampled data. But there are some cases
where FT occurs naturally.

- FT is linear
- A shift in time, introduces a ~~shift~~ ^{slope} in phase
- Vice versa
- A functions increase in periodicity, causes a decrease in frequency. (Scaling property)

~~or~~

conjugate symmetry property

$$[f(t)]^* \Leftrightarrow [\hat{f}(-\omega)]^*$$

$\Rightarrow$ a real function will be conjugate symmetric $\Rightarrow \hat{f}(-\omega) = [\hat{f}(\omega)]^*$
 an imag fn. will be odd symmetric
 $\hookrightarrow \hat{f}(-\omega) = -[\hat{f}(\omega)]^*$

Using this we can write

$$\begin{aligned} F(\omega) &= \frac{1}{2} [\hat{f}(\omega) + \hat{f}(-\omega)]^* \\ G(\omega) &= \frac{1}{2j} [\hat{f}(\omega) - \hat{f}(-\omega)]^* \end{aligned}$$

used in FFT implementation

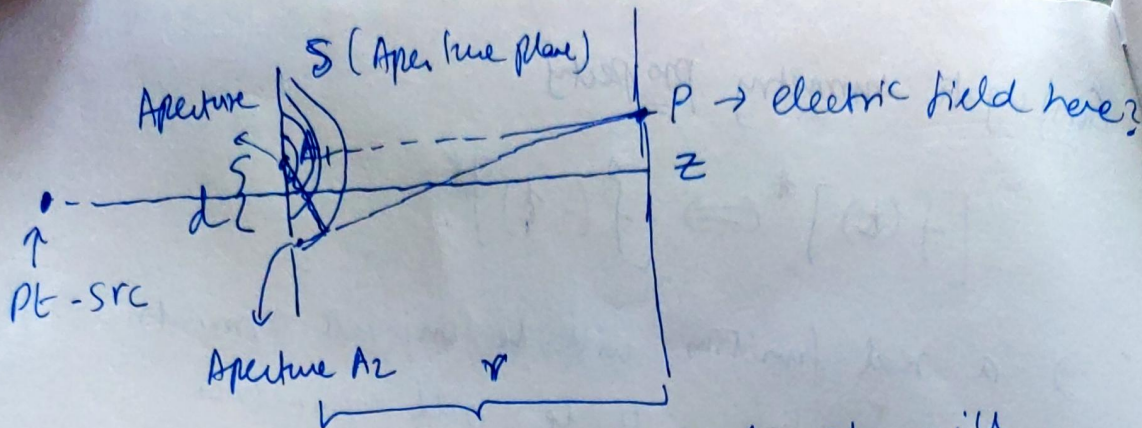
A convolution in time domain is a multiplication in Fourier domain. [filtering]

• There are scenarios in real life where we measure the Fourier transform directly. Happens in optics. To understand this, we will ~~consider~~ go back to the Helmholtz equation, which is a form of the wave equation:

$$\nabla^2 A = -k^2 A$$

$$\Rightarrow \underbrace{(\nabla^2 + k^2) A = 0}_{\downarrow}$$

Time independent form of the wave eqn.



From Huygens's principle, Apertures A_1, A_2 etc. will emit spherical waves, which will add to form the net field at P

Field emitted from area ds at aperture plane S

is :
$$dE = E_0 ds$$

$\hookrightarrow$ original strength of the point source incident on S.

(0)

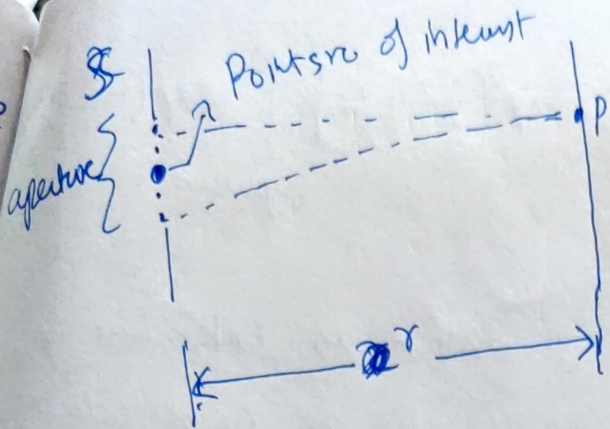
Field due to this at position P on the screen is :

$$dE'' = \frac{dE}{r} = \frac{E_0 \cdot ds}{r}$$

$\therefore E'' = \int_{\text{aperture}} \frac{E_0 ds}{r}$; Phases must be considered too as the field E_0 can be complex.

The electric field can be written using its time varying solution

$$E = E_0 \cdot e^{j2\pi f t}$$



- ϕ at P depends on
- ① Wavelength
 - ② distance, r

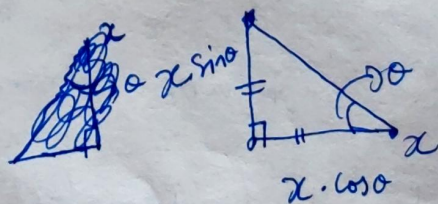
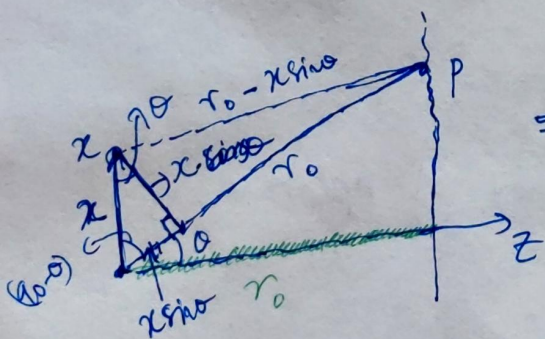
$\therefore$ The phase is $\frac{2\pi}{\lambda} \cdot r \Rightarrow 2\pi \left(\frac{r}{\lambda} \right)$ $\rightarrow r$ in wavelengths

$\Rightarrow$ We can now write dE from on the Plane P, from a small element dx , at a distance x above the aperture's origin as:

$$dE = E_0 \cdot \exp(j2\pi ft) \cdot \exp\left[j\frac{2\pi \cdot r}{\lambda}\right] \cdot dx$$

$\therefore$ The total field at P is then

$$E = E_0 \cdot e^{j2\pi ft} \int_{\text{aperture}} \exp\left(j\frac{2\pi r}{\lambda}\right) \cdot dx$$



$$\theta = 90 - \theta$$

$$\Rightarrow x \cos(90 - \theta) = x \sin \theta$$

$\Rightarrow$ ~~distance~~ distance travelled by a wave from a point x to the point P on the screen is

$$\therefore \underline{r = r_0 - x \sin \theta}$$

$\hookrightarrow$ Plug this in E .

$$E = E_0 \cdot \exp[j2\pi f t] \cdot \exp\left[-j2\pi \frac{r_0}{\lambda}\right] \int_{\text{aperture}} \exp(-j2\pi \frac{x \sin \theta}{\lambda}) dx$$

$\Rightarrow$ ~~Ex~~ If the aperture is written as extending in space from $-d/2$ to $+d/2$ with an Amplitude distribution of varying as $A(x)$, then the integration can be written as:

$$E \propto \int_{-\infty}^{\infty} A(x) \exp\left(-2\pi i \frac{x \sin \theta}{\lambda}\right) dx$$

$$\begin{aligned}
 & p = \frac{\sin \theta}{\lambda} \\
 \Rightarrow E & \propto \underbrace{\int_{-\infty}^{\infty} A(x) e^{-j2\pi p x} \cdot dx}_{\text{Fourier Transform}}
 \end{aligned}$$

Fourier Transform.

$\Rightarrow$ E at ~~the~~ far field is FT of Aperture